

Rean Clive Fernandes

Freiburg im Breisgau, DE

M.Sc. Embedded Engineering (AI Specialization)

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PROFILE

My focus as a Machine Learning Researcher (M.Sc. AI) is on Large Language Models (LLMs) and their application to complex problem-solving. My expertise covers end-to-end ML solutions, from advanced model fine-tuning (Q-LoRa) and algorithm development to building automated data processing and robust evaluation pipelines using Python, PyTorch, and Hugging Face. With a background in Electronics & Communication Engineering and a strong command of MLOps principles, I am keen to contribute to innovative R&D projects and tackle challenging data-driven tasks. I am a quick study with new technologies and scientific domains.

EDUCATION

- M.Sc. Embedded Engineering (Specialization: Artificial Intelligence)** 04/2021 - 09/2024
Albert-Ludwigs-Universität, Freiburg GPA: 2.3
 - Thesis:** Supervised Fine-tuning of Open LLMs for Law: Training and Evaluation for Performance in Legal Examinations (Grade: 1.0)
 - Focus: Deep Learning, Natural Language Processing, Control Theory, Robotics, Optimization.
- B.Engg. Electronics and Communication Engineering** 07/2016 - 07/2020
NMAM Institute of Technology, Karnataka, India GPA: 8.5/10
 - Focus: Advanced Mathematics, Signal Processing, Embedded System Design, Control Theory.

PROFESSIONAL EXPERIENCE

- Research Associate: Legal Large Language Models** Freiburg i. Br., DE
Machine Learning Lab, Albert-Ludwigs-Universität 10/2024 - Present
 - Leading research in applying and optimizing LLMs (e.g., Llama 3 family) for complex legal reasoning tasks.
 - Developing and refining scalable Supervised Fine-tuning (SFT) pipelines and advanced alignment techniques (DPO) using Python, PyTorch, and Hugging Face.
 - Curating high-quality preference datasets for LLM alignment.
 - Designing and analyzing novel decoding methods to enhance LLM performance.
- Research Assistant: Data Curation for Language Model Training** Freiburg i. Br., DE
Machine Learning Lab, Albert-Ludwigs-Universität 04/2024 - 10/2024
 - Engineered automated Python-based data processing workflows for LLM training, including knowledge distillation (Llama 3 70B) enhancing Llama 2 7B performance by 20%.
 - Developed comprehensive benchmarking pipelines for models like Llama and Mistral, showcasing experience with processing diverse unstructured textual data.
 - Curated and cleaned specialized datasets (1500+ examples) to improve LLM training quality.
- Tutor: Foundations of Deep Learning** Freiburg i. Br., DE
Albert-Ludwigs-Universität 10/2023 - 09/2024
 - Mentored students in understanding complex Deep Learning concepts and practical implementations using Python and PyTorch.
 - Developed supplementary materials and facilitated collaborative learning sessions.

PROJECTS

- Master Thesis: Advanced LLM Fine-Tuning & Pipeline Automation for Legal Reasoning** 03/2024 - 09/2024
Led end-to-end development of LLM solutions, achieving 3-4x performance gains by fine-tuning Llama 2 7B (Q-LoRa) on US Bar Exam questions. Engineered automated data distillation (Llama 3 70B, IRAC method) and evaluation pipelines (Python, PyTorch, Hugging Face Transformers on HPC). Created interactive visualizations (Streamlit, Plotly). Grade: 1.0
 - Key Skills: LLM Fine-tuning (Q-LoRa), Data Distillation, Pipeline Automation, Model Evaluation, Prompt Engineering, Python, PyTorch, Hugging Face, HPC, Streamlit, Plotly.
- Smart SLURM Job Handler** Personal Project
Developed a Python-based automated job management tool with SQLite for HPC SLURM clusters, demonstrating practical MLOps skills in pipeline automation and resource optimization.
 - Key Skills: Python, SQLite, SLURM, HPC, Automation, MLOps (Pipeline Automation).
- Non-Linear Model Predictive Control for Crazyflie 2.1** Academic Project (M.Sc.)
Implemented and comparatively analyzed direct collocation and direct multiple shooting NMPC methods for a nano quadcopter. Involved system modeling, simulation, and performance analysis.

- Key Skills: Python, CasADi, Non-Linear Model Predictive Control (NMPC), Control System Design, Numerical Simulation, Data Analysis.

•Optimal Control of a Simulated Furuta Pendulum

Academic Project (M.Sc.)

Designed and simulated MPC controllers (Direct Multiple Shooting, Direct Collocation, Real-Time Iteration) using CasADi on MATLAB for swing-up and stabilization of a Furuta pendulum.

- Key Skills: MATLAB, CasADi, Model Predictive Control (MPC), Numerical Simulation, Control System Design (ODE/DAE modeling).

•Analysis of Trust Region Optimization Methods

Academic Project (M.Sc.)

Implemented and comparatively analyzed various Trust Region sub-problem solution methods (Cauchy Point, Dog-leg, Conjugate Gradient) for non-linear optimization using Python and CasADi.

- Key Skills: Python, CasADi, Numerical Optimization, Mathematical Modelling, Performance Analysis.

•Skill-based Model-based Reinforcement Learning (Research Contribution)

Academic Research Group (M.Sc.)

Contributed to research on combining model-based RL with skill-based approaches for improved sample efficiency and long-horizon task performance (conceptual work and discussions presented in poster format).

- Key Concepts: Reinforcement Learning (RL), Model-Based RL, Variational Autoencoders (VAEs), Skill Extraction.

PUBLICATIONS

•Rean Fernandes, Andre Biedenkapp, Frank Hutter, & Noor Awad

April 2025/Preprint

"A Llama walks into the 'Bar': Efficient Supervised Fine-Tuning for Legal Reasoning in the Multi-state Bar Exam." 

TECHNICAL SKILLS

AI & LLMs: Large Language Model (LLM) Development (Advanced Fine-tuning: Q-LoRa, LoRA concepts, Full Parameter SFT; Model Customization; Robust Evaluation & Experiment Design), LLM Pipelines (End-to-End Development, Automation, Maintenance), Dataset Distillation & Curation (IRAC method), Prompt Engineering (Zero-shot, Few-shot), Transformer Architectures, Text Generation, Alignment Techniques (DPO), Retrieval Augmented Generation (RAG - Conceptual Understanding), NLP, Information Retrieval (IR) Concepts.

Machine Learning Frameworks: PyTorch, Hugging Face (Transformers, Accelerate - conceptual, Datasets, Tokenizers), Scikit-learn, LangChain (Familiarity).

Programming & Data: Python (Expert; clean, efficient, modular code for Data/ML pipelines), SQL (SQLite), Pandas, NumPy, MATLAB, CasADi. Solid understanding of data storage architectures; proficient with concepts of Vector Databases, OLAP, and Lakehouse platforms and ready for practical application.

MLOps & Cloud: MLOps Principles (Model Versioning: Git; Experiment Tracking: Weights & Biases; Automated ML Pipelines: SLURM, Python scripting). Foundational understanding and rapid assimilation of CI/CD for ML, Model Monitoring, and Lifecycle Management. Cloud (AWS): Foundational experience with cloud infrastructure; strong aptitude for rapidly mastering and applying AWS services including SageMaker, Glue, and infrastructure-as-code tools like CDK, Terraform.

Software & Tools: Docker, Git/GitHub, Linux/Unix, Jupyter Notebooks, VS Code, Streamlit, Plotly, SLURM (Admin & Scripting).

Other Technical Areas: Numerical Optimization (Trust Region Methods), Model Predictive Control (MPC), Control System Design & Simulation (ODE/DAE), Robotics (Conceptual), Signal Processing.

Domain Adaptability: Proven ability to apply AI/ML techniques to new domains (e.g., complex legal text analysis). Strong interest in contributing to scientific discovery in diverse fields.

Languages: English (Fluent), German (A2 - Elementary Proficiency)